
SEQUENCES & SERIES

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AZIZ MANVA

AZIZMANVA@GMAIL.COM

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1. BASICS

1.1 Triangular Numbers

A. Basics

The figurate numbers form geometrical figures – triangles, squares, pentagons and others¹. We begin our study of figurate numbers by looking at triangles. The colored green squares in the diagram below form the shape of a triangle. They also form the shape of a staircase.

The numbers in the first n rows of the triangular arrangement above are called triangular numbers. Hence, the first few triangular numbers are:

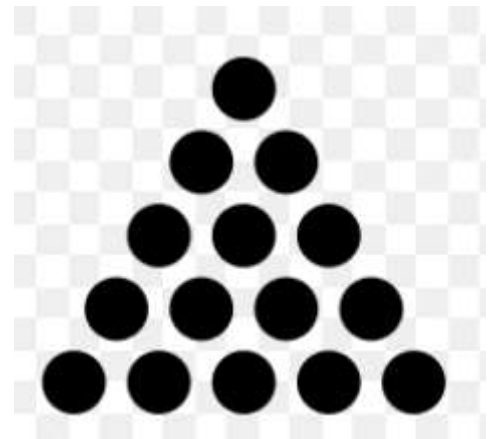
$$\text{First Triangular Number} = n_1 = 1$$

$$\text{Second Triangular Number} = n_2 = 1 + 2 = 3$$

$$\text{Third Triangular Number} = n_3 = 1 + 2 + 3 = 6$$

Example 1.1: Finding Numbers

- Find the sixth triangular number.
- Find the eighth triangular number.
- Find the tenth triangular number using a pattern.



Part A and B

$$\text{Sixth Triangular Number} = 1 + 2 + 3 + 4 + 5 + 6 = 21$$

$$\text{Eighth Triangular Number} = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36$$

Part C

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$$

Add the first number and the last number:

$$= 1 + 10 = 11$$

Add the second number and the second last number

$$= 2 + 9 = 11$$

Add the third number and the third last number

$$= 3 + 8 = 11$$

Add the fourth number and the fourth last number

$$= 4 + 7 = 11$$

Add the fifth number and the fifth last number

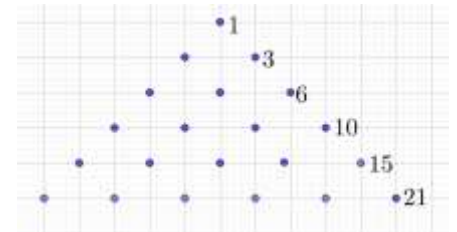
$$= 5 + 6 = 11$$

The number of 11's is the same as the number of pairs that can be made.

And the number of pairs is exactly half of the total number of numbers.

Hence, the total is

$$= 11 \times \frac{10}{2} = 11 \times 5 = 55$$



1.2: Sum of Natural Numbers

We can view the triangular numbers as representing the sum of the first n natural numbers, and find a formula for them.

$$1 + 2 + 3 + \dots + (n - 2) + (n - 1) + n = n \left(\frac{n + 1}{2} \right)$$

¹ See [here](#) for a small discussion on why they are called figurate.

Pair the Numbers as below:

$$1 + 2 + 3 + \dots + (n-2) + (n-1) + n$$

Add the first number and the last number:

$$= 1 + n = n + 1$$

Add the second number and the second last number

$$= 2 + n - 1 = n + 1$$

Add the third number and the third last number

$$= 3 + n - 2 = n + 1$$

Add the fourth number and the fourth last number

$$= 4 + n - 3 = n + 1$$

Each pair adds up to:

$$n + 1$$

And the number of pairs is

$$\frac{n}{2}$$

Final Sum

$$= \underbrace{\frac{n}{2}}_{\text{Number of Pairs}} \times \underbrace{(n+1)}_{\text{Total of each pair}} = \frac{n(n+1)}{2}$$

Example 1.3: Using the Formula

- What is the sum of the first 100 natural numbers?
- Anand counted the natural numbers from 1 to 25. What is the sum of all the numbers he counted?
- What is the sum of the numbers from 20 to 40?
- Compute $\frac{1}{5} + \frac{2}{5} + \dots + \frac{10}{5}$
- What is the sum of the numbers from 51 to 100?
- What is the greatest prime factor of the sum of the arithmetic sequence $1 + 2 + 3 + \dots + 80$?
(MathCounts 2002 School Sprint)
- Find the sum of the first ten even numbers.
- Find the sum of the first twenty multiples of three.

Part A

Substitute $n = 100$ in the formula:

$$\frac{n(n+1)}{2} = \frac{100 \times 101}{2} = 50 \times 101 = 5050$$

Part B

Substitute $n = 25$ in the formula:

$$\frac{n(n+1)}{2} = \frac{25 \times 26}{2} = 25 \times 13 = 250 + 75 = 325$$

Part C

We want the sum

$$20 + 21 + \dots + 40$$

We add and subtract $1 + 2 + \dots + 19$:

$$\underbrace{1 + 2 + \dots + 19}_{\text{Add it here}} + 20 + \dots + 40 - \underbrace{(1 + 2 + \dots + 19)}_{\text{Subtract it here}}$$

Apply the formula for the sum of the first n numbers:

$$= \frac{40 \times 41}{2} - \frac{19 \times 20}{2} = 820 - 190 = 630$$

Part D

$$\frac{1}{5} + \frac{2}{5} + \dots + \frac{10}{5} = \frac{55}{5} = 11$$

Part E

We want the sum:

$$51 + 52 + \dots + 100$$

We add and subtract $1 + 2 + \dots + 50$:

$$1 + 2 + \dots + 100 - (1 + 2 + \dots + 50)$$

Apply the formula for the sum of the first n numbers:

$$= \frac{100 \times 101}{2} - \frac{50 \times 51}{2} = 5050 - 1275 = 3775$$

Part F

$$1 + 2 + \dots + 80 = \frac{80 \times 81}{2} = 40 \times 81 = 2^3 \times 3^4 \times 5$$

$\Rightarrow$ Greatest Prime Factor = 5

Part G

$$2 + 4 + 6 + 8 + \dots + 20$$

Factor out 2 from each term:

$$2(1 + 2 + 3 + \dots + 10) = 2 \left[\frac{10 \times 11}{2} \right] = 110$$

Part H

$$\begin{aligned} 3 + 6 + 9 + 12 + \dots + 60 \\ = 3(1 + 2 + \dots + 20) \\ = 3 \left[\frac{20 \times 21}{2} \right] \\ = 630 \end{aligned}$$

Example 1.4

The natural numbers are written following some rule as 1,3,6,10, ...The tenth number is: (NMTC Primary/Screening 2007/1)

We have the triangular numbers:

$$\text{First Number: } n_1 = 1$$

$$\text{Second Number: } n_2 = 1 + 2 = 3$$

$$\text{Third Number: } n_3 = 1 + 2 + 3 = \frac{3 \times 4}{2} = 6$$

$$\text{Fourth Number: } n_4 = 1 + 2 + 3 + 4 = \frac{4 \times 5}{2} = 10$$

$$\text{Tenth Number: } \frac{10 \times 11}{2} = 5 \times 11 = 55$$

Example 1.5

Look at the rows of numbers shown in the diagram. The first number in the 50th row is: (NMTC Primary/Screening 2004/2)

We can get the blue numbers as the n^{th} triangular number:

$$\text{First Row: } n_1 = 1$$

$$\text{Second Row: } n_2 = 1 + 2 = 3$$

$$\text{Third Row: } n_3 = 1 + 2 + 3 = \frac{3 \times 4}{2} = 6$$

$$\text{Fourth Row: } n_4 = 1 + 2 + 3 + 4 = \frac{4 \times 5}{2} = 10$$

Rather than finding the 50th Row, we find the last number in the 49th Row

$$1 + 2 + \dots + 49 = \frac{49 \times 50}{2} = 49 \times 25 = 1225$$

And hence, the first number in the 50th Row is

$$1225 + 1 = 1226$$

1st Row				1
2nd Row			2	3
3rd Row		4	5	6
4th Row	7	8	9	10

Example 1.6

It takes one block to build a staircase with one step, three blocks to build a staircase with two steps, and six blocks to build a staircase with three steps. The n^{th} step of the staircase needs n blocks. What is the total number of blocks needed to build a staircase with 7 steps?

$$\text{Total Blocks} = 1 + 2 + \dots + 7$$

Substitute $n = 7$ in $\frac{n(n+1)}{2}$:

$$\frac{7 \times 8}{2} = 7 \times 4 = 28$$

Example 1.7

A 12-hour clock strikes once at 1:00, twice at 2:00, etc. How many times does the clock strike in a 24-hour period? (MathCounts 1990 School Sprint)

$$1+2+\dots+12=78$$
$$\text{Ans} = 78 \times 2 = 156$$

Example 1.8

Sean adds up all the even integers from 2 to 500, inclusive. Julie adds up all the integers from 1 to 250, inclusive. What is Sean's sum divided by Julie's sum? (MathCounts 2007 Warm-Up 9)

$$J = 1 + 2 + \dots + 250$$
$$S = 2 + 4 + \dots + 500 = 2(1 + 2 + \dots + 250) = 2J$$
$$S/J = 2J/J = 2$$

Example 1.9

(MathCounts Chapter Sprint 2022/22)

Alva adds up all the positive integers less than 2023 and Beth adds up all the even positive integers less than 2023. What is the remainder when the absolute difference between Alva's and Beth's sums is divided by 1000?

$$\begin{aligned} & (1 + 2 + 3 + \dots + 2022) - (2 + 4 + \dots + 2022) \\ &= \frac{2022(2023)}{2} - 2(1 + 2 + \dots + 1011) \\ &= 1011(2023) - 2\left(\frac{1011(1012)}{2}\right) \\ &= 1011(2023) - 1011(1011) \\ &= 1011^2 \end{aligned}$$

The remainder on dividing by 1000 gives us the last three digits. Hence, we can drop the thousand's digit from the calculations:

$$= 11^2 = 121$$

Example 1.10

Mahadevan conducted a problem-solving session for a group of 18 primary class students. Seeing the graded performance, he distributed packets of biscuits to all the students.

The first student got 1 packet plus $\frac{9}{19}$ of 1 packet.

The second student got 2 packets plus $\frac{9}{19}$ of 2 packets.

The third student got 3 packets plus $\frac{9}{19}$ of 3 packets, and so on.

The total number of packets of biscuits distributed by Mahadevan is: (NMTC Primary/Screening/2016/13)

The total number of packets is:

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$$\underbrace{\left(1 + \frac{9}{19} \cdot 1\right)}_{1st\ Student} + \underbrace{\left(2 + \frac{9}{19} \cdot 2\right)}_{2nd\ Student} + \cdots + \underbrace{\left(18 + \frac{9}{19} \cdot 18\right)}_{18th\ Student}$$

Factor 1 from the first term, 2 from the second term and so on:

$$1\left(1 + \frac{9}{19}\right) + 2\left(1 + \frac{9}{19}\right) + \cdots + 18\left(1 + \frac{9}{19}\right)$$

Factor $1 + \frac{9}{19}$ from each term:

$$\left(1 + \frac{9}{19}\right)(1 + 2 + \cdots + 18)$$

Use the formula for the sum of triangular numbers:

$$\left(1 + \frac{9}{19}\right)\left(\frac{18 \cdot 19}{2}\right) = \left(\frac{28}{19}\right)\left(\frac{18 \cdot 19}{2}\right) = (28)(9) = 252$$

B. Back Calculations

Example 1.11: Back Calculations

If the sum $1 + 2 + 3 + \cdots + K$ is a perfect square N^2 and if N is less than 100, then the possible values for K are: (AHSME 1956/36)

- A. Only 1
- B. 1 and 8
- C. Only 8
- D. 8 and 49
- E. 1, 8 and 49

$$K = 1 \Rightarrow 1 + 2 + 3 + \cdots + K = 1 = 1^2 \Rightarrow \text{Valid}$$

$$K = 8 \Rightarrow 1 + 2 + 3 + \cdots + K = 36 = 6^2 \Rightarrow \text{Valid}$$

$$K = 49 \Rightarrow 1 + 2 + 3 + \cdots + K = 1225 = 35^2 \Rightarrow \text{Valid}$$

Option E is correct

Example 1.12

If 75 is written as the sum of 10 consecutive natural numbers, the maximum of the numbers is: (NMTC Primary-III/Screening/36)

The sum of the numbers from 1 to 10 is:

$$1 + 2 + \cdots + 10 = \frac{10 \times 11}{2} = 5 \times 11 = 55$$

If we increase every number by 1, then we get, and the overall total increases by 10:

$$2 + 3 + \cdots + 11 = 65$$

If we add 1 to every number one more time, then the total again increases by 10:

$$3 + 4 + \cdots + 12 = 75$$

Hence, the maximum of the numbers is:

$$12$$

Example 1.13

The number 21 is the sum of two consecutive integers 10 and 11. What is the largest number of positive,

consecutive integers whose sum is 21? (**MathCounts 2008 State Countdown**)

$$1+2+\dots+6=21$$

$$1+2+\dots+7=28$$

$$\text{Ans}=6$$

Example 1.14

When reading a book, Charlie made a list by writing down the page number of the last page he finished reading at the end of each day. (He always finished reading a page that he started.) His mom thought his list indicated the amount of pages he had read on each day. At the end of the 8th day of reading, she added the numbers on his list and thought Charlie had read 432 pages. If Charlie started reading the book on page one, and he read the same amount of pages each day of this eight-day period, how many pages did he actually read by the end of the day? (**MathCounts 2006 Chapter Sprint**)

Let the number of pages read each day be a . Then, Charlie will write

$$a, 2a, 3a, \dots, 8a$$

His mom will add:

$$a + 2a + \dots + 8a = 432$$

Take a common:

$$a(1 + 2 + \dots + 8) = 432 \Rightarrow a\left(\frac{8 \times 9}{2}\right) = 432$$
$$36a = 432$$

The actual number of pages will be:

$$8a = \left(\frac{432}{36}\right) \times 8 = 96$$

Example 1.15

An airplane climbs 100 feet during the first second after takeoff. In each succeeding second it climbs 100 feet more than it climbed during the previous second. How many seconds does it take for the plane to reach an altitude of 12,000 feet above its takeoff height? (**MathCounts 2007 School Sprint**)

Suppose the flight takes n seconds:

$$100 + 200 + \dots + 100n = 12000$$

$$100(1 + 2 + 3 + \dots + n) = 12000$$

$$\frac{n(n+1)}{2} = 120$$

$$n(n+1) = 240$$

Factor pairs of 240 are:

$$(1,240), (2,120), (3,80), (4,60), (5,48),$$

$$(6,40), (8,30), (10,24), (12,20), (15,16)$$

$$n = 15$$

You can also do this using quadratics:

$$n(n+1) = 240 \Rightarrow n^2 + n - 240 = 0$$

Example 1.16

The pages of a book are numbered 1 through n . When the page numbers of the book were added, one of the page numbers was mistakenly added twice, resulting in an incorrect sum of 1986. What was the number of the page that was added twice? (**AIME 1986/6**)

$$\begin{aligned}n = 10 &\Rightarrow \frac{n(n+1)}{2} = \frac{10 \times 11}{2} = 55 \\n = 40 &\Rightarrow \frac{n(n+1)}{2} = \frac{40 \times 41}{2} = 820 \\n = 60 &\Rightarrow \frac{n(n+1)}{2} = \frac{60 \times 61}{2} = 1830 \\n = 61 &\Rightarrow \frac{n(n+1)}{2} = \frac{61 \times 62}{2} = 1892 \\n = 62 &\Rightarrow \frac{n(n+1)}{2} = \frac{62 \times 63}{2} = 1953 \\n = 63 &\Rightarrow \frac{n(n+1)}{2} = \frac{63 \times 64}{2} = 2016\end{aligned}$$

Hence, the number of pages in the book are:

$$62$$

The wrong total is 1986. The actual total is 1953. Hence, it is more by:

$$\begin{aligned}1986 - 1953 &= 33 \\ \frac{66}{2} &= 33\end{aligned}$$

Example 1.17

Starting from her tenth birthday, Dawn reads 1 page on the first day, 2 pages the second day, 3 pages the third day, etc? What is the least number of days that she can read so that her total pages read are a multiple of 56?

Let the number of days she reads be n . Then for some natural number k :

$$\frac{n(n+1)}{2} = 56k \Rightarrow n(n+1) = 112k$$

Prime Factorize 112:

$$112 = 2^4 \times 7$$

The factors of 2 can be split across n and $n+1$, but one of the two must be a multiple of 7.

$$7 \times 8 = 56 \Rightarrow \text{Too Small}$$

Check with 14:

$$14 \times 15, 14 \times 13 \Rightarrow \text{Both 15 and 13 are odd} \Rightarrow \text{Does not work}$$

(In fact, if the multiple of 7 is even, the number before and after will be odd. So, we do not need to check unless we reach $7 \times 16 = 112$)

Check with 21:

$$21 \times 22 \Rightarrow \text{Only 1 factor of 2}$$

$$21 \times 20 \Rightarrow \text{Only 2 factors of 2}$$

Check with 35:

$$35 \times 36 \Rightarrow \text{Only 2 factors of 2}$$

$$35 \times 34 \Rightarrow \text{Only 1 factor of 2}$$

Check with 49:

$$49 \times 50 \Rightarrow \text{Only 1 factors of 2}$$

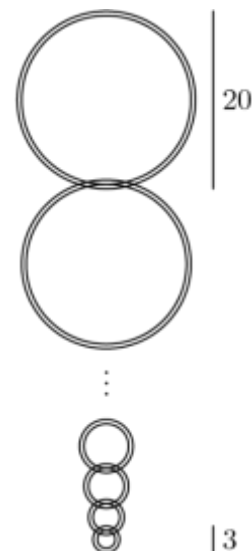
$$49 \times 48 \Rightarrow 4 \text{ Factors of 2} \Rightarrow \text{Works}$$

Hence,

$$n = 48$$

Example 1.18

A number of linked rings, each 1 cm thick, are hanging on a peg. The top ring has an outside diameter of 20 cm. The outside diameter of each of the other rings is 1 cm less than that of the ring above it. The bottom ring has an outside diameter of 3 cm. What is the distance, in cm, from the top of the top ring to the bottom of the bottom ring? (AMC 10A 2006/14)



Method II

If the rings were *not* linked, then you could put next to each other, and the total distance would be:

$$\begin{aligned} & 3 + 4 + \dots + 20 \\ &= 1 + 2 + \dots + 20 - 3 \\ &= \frac{20 \times 21}{2} - 3 \\ &= 210 - 3 = 207 \end{aligned}$$

Since the rings are linked, we must subtract the overlap. There are:

$$\begin{aligned} 20 - 2 &= 18 \text{ Rings} \\ 18 - 1 &= 17 \text{ Overlaps} \\ \text{Overlap Distance} &= 2 \times 17 = 34 \end{aligned}$$

And hence the final answer is:

$$207 - 34 = 173$$

Method I

The distance from the top of the largest, to its bottom, is simply:

$$20$$

The bottom of the top ring is 2 cm below the top of the 2nd ring. Hence, its distance from the bottom of the 2nd ring is

$$19 - 2 = 17$$

Similarly, the bottom of the 2nd ring is 2 cm below the top of the 3rd ring, and hence its distance from the bottom of the 3rd ring is:

$$18 - 2 = 16$$

We continue this pattern till we reach the last ring, getting a total of:

$$\begin{aligned} & 20 + 17 + 16 + \dots + 1 \\ &= 20 + \frac{17 \times 18}{2} = 20 + 153 = 173 \end{aligned}$$

C. Applications

Example 1.19: Positions

The positive integers are arranged in rows and columns as shown. More rows continue to list the positive integers in order, with each new row containing one more integer than the previous row. How many integers less than 2000 are in the column that contains the number 2000?

(Gauss Grade 7 2012/25)

Row 1	1					
Row 2	2	3				
Row 3	4	5	6			
Row 4	7	8	9	10		
Row 5	11	12	13	14	15	
Row 6	16	17	18	19	20	21

The number at the rightmost end are:

$1, 3, 6, 10 \Rightarrow \text{Triangular Numbers}$

The rightmost number in the row in which 2000 occur (using trial and error) is:

$$\frac{63 \times 64}{2} = 2016$$

We start counting the numbers in the column above 2016, and going left:

$(2016, 0), (2015, 1), (2014, 2), \dots, (2016 - n, n)$

Since

$$2016 = 2000 - 16 \Rightarrow n = 16$$

There are

$16 \text{ Numbers above } 2000$

Example 1.20: Average

Find the average of the positive numbers less than 100 which are multiples of 3.

The numbers are:

$$3 + 6 + 9 + \dots + 99$$

Factor 3 from each term

$$= 3(1 + 2 + 3 + \dots + 33)$$

The sum of the above expression is:

$$= 3 \left(\frac{33 \cdot 34}{2} \right)$$

The list $\{1, 2, \dots, 33\}$ has 33 numbers and hence to find the average, we divide by 33:

$$= 3 \left(\frac{33 \cdot 34}{2} \right) \times \frac{1}{33} = 3 \left(\frac{34}{2} \right) = 3(17) = 51$$

Example 1.21: Average

Find the average of the first n multiples of the natural number m .

The numbers are:

$$m + 2m + 3m + \dots + nm$$

Factor m from each term

$$\begin{aligned} & m(1 + 2 + 3 + \dots + n) \\ & = m(1 + 2 + 3 + \dots + n) \end{aligned}$$

The sum of the above expression is:

$$= m \left(\frac{n(n+1)}{2} \right)$$

The list $\{1, 2, \dots, n\}$ has n numbers and hence to find the average, we divide by n :

$$= m \left(\frac{n(n+1)}{2} \right) \times \frac{1}{n} = m \left(\frac{n+1}{2} \right)$$

1.2 Square Numbers

A. Square Numbers

1.22: Sum of Odd Numbers

The sum of the first n odd numbers:

$$\underbrace{1 + 3 + 5 + \cdots}_{n \text{ terms}} = n^2$$

1		1	3		1	3	5		1	3	5	7		1	3	5	7	9		1	3	5	7	9	11

$$1 + 3 = 4 = 2^2$$

$$1 + 3 + 5 = 9 = 3^2$$

$$1 + 3 + 5 + 7 = 16 = 4^2$$

$$1 + 3 + 5 + 7 + 9 = 25 = 5^2$$

$$1 + 3 + 5 + 7 + 9 + 11 = 36 = 6^2$$

Example 1.23

Find the sum of the numbers in each part below. Write each number as the square of a natural number.

- A. $1 + 3 + 5$
- B. $1 + 3 + 5 + 7$
- C. $1 + 3 + 5 + 7 + 9$
- D. $1 + 3 + 5 + 7 + 9 + 11$
- E. $1 + 3 + 5 + \cdots + 101$
- F. $(-39) + (-37) + \cdots + (-1)$ (MathCounts 1990 School Sprint)
- G. $-1 - 49 - 3 - 47 - \cdots - 23 - 25$

$$\underbrace{1 + 3 + 5}_{3 \text{ Numbers}} \Rightarrow n = 3 \Rightarrow n^2 = 3^2 = 9$$

$$\underbrace{1 + 3 + 5 + 7}_{4 \text{ Numbers}} \Rightarrow n = 4 \Rightarrow n^2 = 4^2 = 16$$

$$\underbrace{1 + 3 + 5 + 7 + 9}_{5 \text{ Numbers}} \Rightarrow n = 5 \Rightarrow n^2 = 5^2 = 25$$

$$1 + 3 + 5 + 7 + 9 + 11 \Rightarrow n = \frac{11 + 1}{2} = 6 \Rightarrow n^2 = 6^2 = 36$$

Using sum of odd numbers:

$$-(1 + 3 + 5 + 7 + \cdots + 39) = -\left(\frac{39 + 1}{2}\right)^2 = -20^2 = -400$$

Rearrange to get:

$$-1 - 3 - 5 - \cdots - 47 - 49 = -\left(\frac{49 + 1}{2}\right)^2 = -25^2 = -625$$

Example 1.24

- A. Find the sum to 10 terms of $1 + 3 + 5 + \cdots$. Also find the last term added in the above sequence?

- B. Sheela added all the single digit odd numbers. Find the total she got.
C. Cho added all the two-digit odd numbers. Find Cho's total
D. Find the sum of all odd three-digit integers.
E. What is the sum of the odd integers between 46 and 64? (**MathCounts 2001 National Countdown**)

Part A

$$\underbrace{1 + 3 + 5 + \dots}_{10 \text{ Terms}} = 100 = 10^2$$

Since there are 10 terms, the last number will be:

$$\begin{aligned} 10\text{th Odd No.} &= 10\text{th Even No.} - 1 \\ &= 2(10) - 1 = 20 - 1 = 19 \end{aligned}$$

Part B

We want:

Part C

$$\underbrace{1 + 3 + 5 + 7 + 9}_{5 \text{ Numbers}} \Rightarrow n = 5 \Rightarrow n^2 = 5^2 = 25$$

Part D

$$11 + 13 + \dots + 99$$

Add and Subtract $1 + 3 + 5 + 7 + 9$:

$$\begin{aligned} &= \mathbf{1 + 3 + \dots + 9} + 11 + 13 + \dots + 99 \\ &\quad - (1 + 3 + \dots + 9) \\ &= 2500 - 25 = 2475 \end{aligned}$$

Part E

$$\begin{aligned} &101 + 103 + \dots + 999 \\ &= \underbrace{1 + 3 + \dots + 99}_{\text{Add}} + 101 + 103 + \dots + 999 \\ &\quad - \left(\underbrace{1 + 3 + \dots + 99}_{\text{Subtract}} \right) \\ &= 500^2 - 50^2 = 250000 - 2500 = 247500 \end{aligned}$$

Part F

There are $\frac{63+1}{2} = \frac{64}{2} = 32$ odd integers upto 63. And we do not want the odd integers from 1 to 45, which are

$$\frac{45 + 1}{2} = \frac{46}{2} = 23$$

Use Complementary Counting:

$$32^2 - 23^2 = 1024 - 529 = 495$$

We can also do this using arithmetic series:

$$n \times \left(\frac{\text{First} + \text{Last}}{2} \right) = 9 \left(\frac{47 + 63}{2} \right) = 9 \times 55 = 495$$

Example 1.25

Andy added all the odd numbers from 1 to 100. Find the total he got.

$$n^{\text{th}} \text{ odd number is } 2n - 1$$

The last odd number for Andy was

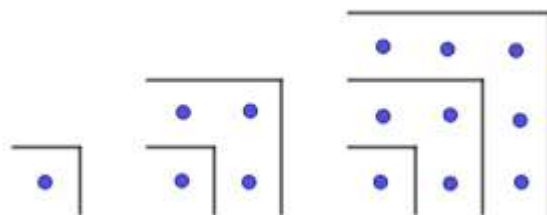
$$\begin{aligned} 2n - 1 &= 99 \\ 2n &= 100 \\ n &= 50 \end{aligned}$$

Hence, there are 50 odd numbers that Andy counted:

$$n^2 = 50^2 = 2500$$

Example 1.26

Look at the following dot diagram. This pattern continues. The value of $1 + 3 + 5 + \dots$ upto 100 terms is the number of dots shown in which pattern? What is the sum $1 + 3 + 5 + \dots$ upto 100 terms (**NMTC Primary/Screening 2004/01, Adapted**)



$$\text{Pattern 1} = 1 \text{ Dot} = 1^2$$

$$\text{Pattern 2} = 4 \text{ Dots} = 2^2$$

$$\text{Pattern 3} = 9 \text{ Dots} = 3^2$$

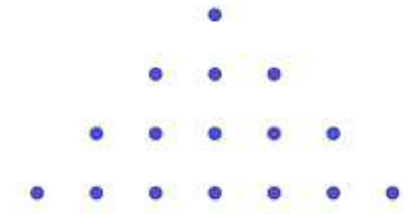
$$n^{\text{th}} \text{ pattern} = n^2 \text{ dots}$$

Part A

100th Pattern

Part B

$$100^2 = 10,000$$



Example 1.27

The total number of dots in the first 100 rows:
 (NMTC Primary/Screening 2005/09)

The number of dots is:

$$1 + 3 = 4 = 2^2$$

$$1 + 3 + 5 = 9 = 3^2$$

$$1 + 3 + 5 + 7 = 16 = 4^2$$

$$1 + 3 + 5 + 7 + 9 = 25 = 5^2$$

Hence, the number of dots in 100 rows is

$$\underbrace{1 + 3 + 5 + \dots}_{100 \text{ Terms}} = 100^2 = 10,000$$

Example 1.28

The natural numbers are written in the following form:

$$1$$

$$2 \ 3 \ 4$$

$$9 \ 8 \ 7 \ 6 \ 5$$

$$10 \ 11 \ 12 \ 13 \ 14 \ 15 \ 16$$

The number 2010 is which position in which row? (NMTC Primary/Screening 2010/10)

To get the row number, we need to first check which two perfect squares does 2010 lie between? After some trial and error, we find that:

$$1936 = 44^2 < 2010 < 45^2 = 2025$$

$$1$$

$$2 \ 3 \ 4 = 2^2$$

$$3^2 = 9 \ 8 \ 7 \ 6 \ 5$$

$$10 \ 11 \ 12 \ 13 \ 14 \ 15 \ 16 = 4^2$$

Note that odd perfect squares are on the left, and even perfect squares are on the right.
 The number that we want is between 44^2 and 45^2 . Hence, our row will look like this:

$$\dots 1936$$

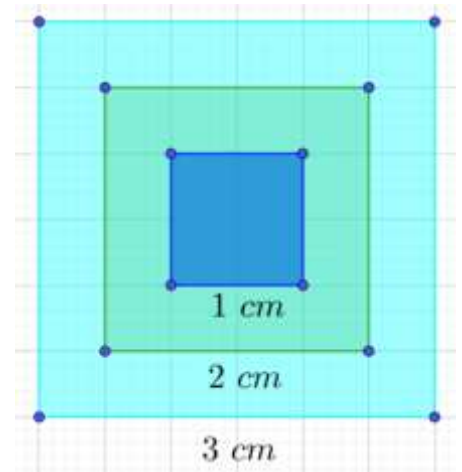
$$2025 \ 2024 \dots 1939 \ 1938 \ 1937$$

2025 is $2025 - 2024 = 1\text{st Number from the Left}$
2024 is $2025 - 2023 = 2\text{nd Number from the Left}$
2010 is $2025 - 2009 = 16\text{th Number from the Left}$

Example 1.29

Candice is drawing squares with the same intersection point for their diagonals. The first square has a side length of 1 cm, the second a side length of 2 units, the third 3 units and so on. She colors the first(smallest) square *dark blue*. Then, she colors the area outside the first square, and inside the second square *green*. She colors the area outside the second square, but inside the third square *light blue*. The diagram shows the first three squares. She continues this process, using a different color each time.

- Let t_n be the area occupied by the n^{th} color that she uses. Find the sequence $t_1, t_2, t_3, \dots$
- Find the sum to 100 terms of $t_1 + t_3 + t_5 + \dots$



Area of 1st Square = 1
Area of 2nd Square = 4
Area of 3rd Square = 9
Area of 4th Square = 16

$$\begin{aligned} t_1 &= 4 - 1 = 3 \\ t_2 &= 9 - 4 = 5 \\ t_3 &= 16 - 9 = 7 \end{aligned}$$

Example 1.30

Ali has an elephant who loves peanuts. He gives his elephant one peanut on the 1st of the month, 3 peanuts on the second day of the month, 5 peanuts on the third day of the month, and so on. When the month changes, he gives his elephant 1 peanut on the first day of the new month. He must feed his elephant the full quota of peanuts every day.

- How many peanuts will his elephant eat in February in a non-leap year?
- Ali has 400 peanuts in stock on 1st Feb before feeding any peanuts to his elephant. He has a just-in-time peanut order policy. They are delivered only when needed. On which day in February will he get his first delivery.
- How many more peanuts will Ali's elephant eat in March as compared to Feb.

Part A

$$1 + 3 + \dots = 28^2 = 784$$

Part B

$$400 = 20^2 \Rightarrow 20 \text{ Days} \Rightarrow \text{Delivery will happen on 21st Day}$$

Part C

$$31^2 - 28^2 = 961 - 784 = 177 \text{ Peanuts}$$

Example 1.31

A book has 100 pages. Strangely, the author insists on giving only odd numbers to the pages

$$\underbrace{1}_{1\text{st Page}}, \underbrace{3}_{2\text{nd Page}}, \dots, \underbrace{199}_{100\text{th Page}}$$

If the sum of the numbers on the pages can be written as $p^a q^b$, where p and q are prime numbers, and a and b

are positive integers, find

$$p + q + a + b$$

There are 100 pages, which will be numbered

$$1 + 3 + 5 + \dots = 100^2 = 10^4 = 2^4 \times 5^4 \Rightarrow 2 + 5 + 4 + 4 = 15$$

Example 1.32

A grocer makes a display of cans in which the top row has one can and each lower row has two more cans than the row above it.

- If the display contains 100 cans, how many rows does it contain? (AMC12B 2004/8)
- What is the number of cans in the bottom row?

$$1 + 3 + 5 + \dots = 100 = 10^2 \Rightarrow 10 \text{ Numbers} \Rightarrow 10 \text{ Rows}$$

$$\text{No. of Cans} = 10\text{th Odd No.} = 2(10) - 1 = 20 - 1 = 19$$

Example 1.33

The sum of 5 consecutive even integers is 4 less than the sum of the first 8 consecutive odd counting numbers. What is the smallest of the even integers? (AMC 10B 2003/3)

$$(1 + 3 + 5 + \dots + 15) = 64$$

$$64 - 4 = 60$$

$$\frac{60}{5} = 12$$

Numbers are 8, 10, 12, 14, 16

Smallest is 8

Example 1.34

Anisha arranged her Lego blocks in groups of size x , starting from the smallest possible group, and increasing sequentially, except that no group was divisible by 2. She had none left over. The blocks were then counted to be 1370. If a single group was counted twice, which group was it?

$$1 + 3 + 5 + \dots = x^2$$

Hence, we need the largest square which is smaller than 1370, which is:

$$37^2 = 1369$$

$$1370 - 1369 = 1$$

Hence, the group of blocks that had a single block in it was counted twice.

Example 1.35

The sequence

$$1, 2, 1, 2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 2, 1, 2, \dots$$

consists of 1's separated by blocks of 2's with n 2's in the n^{th} block. The sum of the first 1234 terms of this sequence is (AHSME 1996/24)

	1 st Block	2 nd Block	3 rd Block	4 th Block	5 th Block				
	1	2,1	2,2,1	2,2,2,1	2,2,2,2,1				
No. of Numbers	1	2	3	4	5	.	.	.	Triangular Numbers

Total of those Numbers	1	3	5	7	9	.	.	.	Square Numbers
------------------------	---	---	---	---	---	---	---	---	----------------

The numbers of numbers in each block is given by the triangular numbers. Hence, we will be able to have:

$$n = 49 \Rightarrow \frac{n(n+1)}{2} = \frac{49 \times 50}{2} = 1225 \Rightarrow 49 \text{ Complete Blocks}$$

The total of those 49 complete blocks will be given by:

$$49^2 = 2401$$

But we need the total of

$$1225 + 9 = 1234 \text{ Terms}$$

After the 49th block, there will be 49 consecutive 2's. But we only need nine of those to get the total to 1234 terms.

So we need to add:

$$2401 + (9 \times 2) = 2401 + 18 = 2419$$

B. Others

Example 1.36

Pentagonal Numbers

Hexagonal Numbers

Centered Hexagonal Numbers

Example 1.37

In the triangle shown, the first diagonal line, 1, 2, 3, 4, ... , begins at 1 and each number after the first is one larger than the previous number. The second diagonal line, 2, 4, 6, 8, ... begins at 2 and each number after the first is two larger than the previous number. The n th diagonal line begins at n and each number after the first is n larger than the previous number. In which horizontal row does the number 2016 first appear? (Gauss Grade 7 2016/25)

A number from the n^{th} diagonal line appears in the m^{th} horizontal row when:

$$\underbrace{n \times m = 2016}_{\text{Number}} \Leftrightarrow \underbrace{n + m - 1}_{\text{Row Number}}$$

To minimize $n + m - 1$, we minimize $n + m$, which is itself minimized by having n and m as close to each other as possible.

We can list the factor pairs of 2016, and the factor pair with minimum difference between the factors is:

$$42 \times 48 = 2016$$

$$n + m - 1 = 42 + 48 - 1 = 89$$

1.3 Common Sequences and Series

A. Defined Sequences

Example 1.38

Two sequences S_1 and S_2 are as under:

$$S_1: \frac{2}{1}, \frac{4}{3}, \frac{6}{5}, \dots$$

$$S_2: \frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \dots$$

The n^{th} term of S_1 is $S_1: \frac{2n}{2n-1}$ and the n^{th} term of S_2 is $\frac{2n-1}{2n}$. The value of the difference between the 2012th terms of S_1 and S_2 is: **(NMTC Primary/Screening 2012/A/5)**

Method I: Substitute first and simplify later:

$$\begin{aligned} 2012^{\text{th}} \text{ term of } S_1 &= \frac{2n}{2n-1} = \frac{2 \times 2012}{2 \times 2012 - 1} = \frac{4024}{4023} \\ 2012^{\text{th}} \text{ term of } S_2 &= \frac{2n-1}{2n} = \frac{2 \times 2012 - 1}{2 \times 2012} = \frac{4023}{4024} \end{aligned}$$

$$\frac{4024}{4023} - \frac{4023}{4024} = \frac{4024^2 - 4023^2}{4024 \times 4023}$$

Factor the numerator using the property $a^2 - b^2 = (a + b)(a - b)$:

$$= \frac{(4024 + 4023)(4024 - 4023)}{4024 \times 4023} = \frac{(4024 + 4023)(1)}{4024 \times 4023} = \frac{8047}{4024 \times 4023}$$

Method II: Simplify first and substitute later

$$\frac{2n}{2n-1} - \frac{2n-1}{2n} = \frac{(2n)^2 - (2n-1)^2}{2n(2n-1)} = \frac{4n^2 - (4n^2 - 4n + 1)}{2n(2n-1)} = \frac{4n-1}{2n(2n-1)}$$

Substitute $n = 2012$:

$$\frac{2012(4) - 1}{(2012 \times 2)(2012 \times 2 - 1)} = \frac{8048 - 1}{(4024)(4024 - 1)} = \frac{8047}{(4024)(4023)}$$

B. Prime Numbers

Example 1.39

Find the 27th term in the sequence:

2,3,5,7,11,13,17, ...

Let's apply common differences:

2,3,5,7,11,13,17, ...
1,2,2,4,2,4

The common differences don't have a clear pattern from which to make a rule.

So, we look for another rule. The above sequence consists of prime numbers, starting from 2.

There are 25 prime numbers less than 100.

$$\therefore 25^{\text{th}} \text{ Prime Number} = 97$$

$$\therefore 26^{\text{th}} \text{ Prime Number} = 101$$

$$\therefore 27^{\text{th}} \text{ Prime Number} = 103$$

Example 1.40

The 25th term in the sequence:

(1,2), (2,3), (3,5), (4,7), (5,11), (6,13), ...

(NMTC Primary/Screening 2004/03)

We can think of this as sequences put together:

First Sequence: 1,2,3,4,5,6,7

Second Sequence: 2,3,5,7,11,13,17, ...

The first number in each bracket is just the term number. Hence, for the 25th term, it will be just
25

The second term in each bracket is a prime number, starting from 2, and then going to the next largest prime, etc. Hence, for the 25th, we want the 25th prime number, which is:
97

Our final answer is:

(25,97)

C. Composite Numbers

Example 1.41

Here is a sequence of composite numbers having only one prime factor, written in ascending order 4,8,9,16,25,27,32, The 15th number of this sequence is:
(NMTC Primary/Screening 2012/B/2)

You cannot have prime numbers:

2,3,5,7, ... $\Rightarrow$ Ruled Out

You cannot have 1:

$\Rightarrow$ Neither prime nor composite.

The only remaining numbers are those which are a power of a single prime.

Powers of 2	4	8	16	32	64	128
Powers of 3	9	27	81	243		
Powers of 5	25	125	625			
Powers of 7	49	343				
Powers of 11	121					
Powers of 13	169					

4,8,9,16,25,27,32,49,64,81,121,125,128,169,243

D. Double Sequences

Example 1.42

Find the next four numbers in the sequence: 11,88,16,80,21,72, ... (NMTC Primary-III/11, Adapted)

This is a double sequence. The odd terms have the rule *Add 5*.

11,16,21,26,31

The even terms have the rule *Subtract 8*.

88,80,72,64,56

The final sequence is:

11,88,16,80,21,72, $\underbrace{26}_{21+5}$, $\underbrace{64}_{72-8}$, $\underbrace{31}_{26+5}$, $\underbrace{56}_{64-8}$

Example 1.43

In the sequence 1,4,3,6,5,8,7,10 ... we have $t_{2n-1} = 2n - 1$ and $t_{2n} = t_{2n-1} + 3$ i.e. every odd term is that odd number and the next even term is 3 more than the previous odd term. If $t_m = 2010$, then $m =$ (NMTC Primary/Screening 2010/14)

We can think of this as a double sequence:

Odd Terms: 1,3,5,7, ...
Even Terms: $\underbrace{4}_{2^{nd} \text{ Term}}$, $\underbrace{6}_{4^{th} \text{ Term}}$, $\underbrace{8}_{6^{th} \text{ Term}}$, $\underbrace{10}_{8^{th} \text{ Term}}$, ..., $\underbrace{2010}_{2008^{th} \text{ Term}}$

$$t_{2n} = t_{2n-1} + 3 = 2n - 1 + 3 = 2n + 2$$

Hence, the value of the term is 2 more than the term number.

E. Squares of Natural Numbers

Example 1.44

- What percent of the first ten thousand natural numbers are perfect squares?
- In the sequence 1, 2, 3, 4, 5, ..., 100000000 the percentage of square numbers is: (NMTC Primary/Screening 2009/11)

Part A

The squares upto $10,000 = 100^2$ are

$$\frac{100}{10000} \times 100 = 1\%$$

Part B

The squares upto $100,000,000 = 10,000^2$ are:

$$1^2, 2^2, 3^2, \dots, 10000^2 \Rightarrow 10,000 \text{ Numbers}$$

And the percentage is:

$$\frac{10,000}{100,000,000} \times 100 = \frac{1}{100} \% = 0.01\%$$

Example 1.45

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Using the formula above, find the sum of the squares:

- of the first seven natural numbers.
- of the numbers $\{-10, -9, -8, \dots, 9, 10\}$
- of the natural numbers from 4 to 20 (Calculator Allowed)

Part A

Substitute $n = 7$ in $\frac{n(n+1)(2n+1)}{6}$ giving:

$$\frac{n(n+1)(2n+1)}{6} = \frac{(7)(8)(15)}{6} = 7 \times 4 \times 5 = 140$$

Part B

Break this into two parts

$$\begin{aligned} &(-10)^2 + (-9)^2 + \dots + (-1)^2 \\ &(10)^2 + (9)^2 + \dots + (1)^2 \end{aligned}$$

Note that each part will have the same sum since

$$(-x)^2 = x^2$$

So, we find

$$2 \left(\frac{(10)(11)(21)}{6} \right) = 770$$

Part C

$$4^2 + 5^2 + \dots + 20^2$$

Add and subtract $1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$

$$1^2 + 2^2 + \dots + 20^2 - 14 = \frac{20 \times 21 \times 41}{6} - 14 = 2870 - 14 = 2856$$

Example 1.46

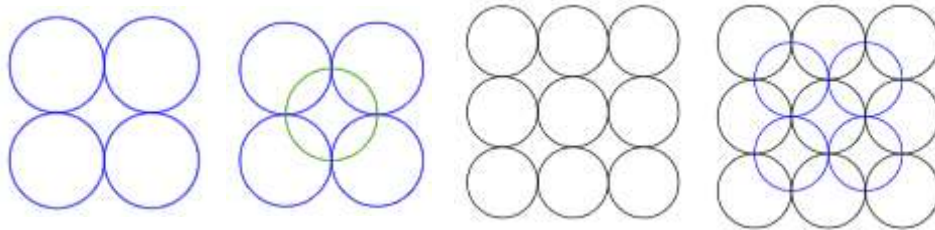
The sum of the squares of the first n positive integers is given by the expression $\frac{n(n+c)(2n+k)}{6}$, if c and k are, respectively: (AHSME 1957/30)

$$c = k = 1$$

Example 1.47

Cannonballs are stacked in a pyramid format in three dimensions as shown. Each cannonball is stacked in the dip or gap between four cannonballs in the layer below it.²

- Find the number of cannonballs in each of the first four rows starting from the topmost row, which has a single cannonball, as can be seen in the photo.
- Find the number of cannonballs in rows 1 to n . Write it as a series. Also, write it as a single formula.
- If the number of cannonballs in n rows is 55, find n .



Part A

The topmost layer has to be, as seen in the photo,

1 Cannonball

Each cannonball is stacked on four cannonballs. Hence, the second layer must be:

4 Cannonballs

If you draw out the number of cannonballs needed to place 4 cannonballs in a 2 by 2 square, you need a three by three square, giving us

9 Cannonballs

If you draw out the number of cannonballs needed to place 9 cannonballs in a 4 by 4 square, you need a three by three square, giving us

16 Cannonballs

Part B

$$\underbrace{1^2 + 2^2 + \dots + n^2}_{\text{Series Version}} = \underbrace{\frac{n(n+1)(2n+1)}{6}}_{\text{Formula}}$$

² The above question only asks for the number of spheres. [Packing maximum spheres in a given space](#) is a much more difficult problem, and is a current research topic in Maths.

Part C

Example 1.48

Oranges are stacked in a three-dimensional pyramid with a single orange stacked in the gap between four oranges in the layer below it. The topmost layer has a single orange. The oranges are then rearranged into a single layer in the shape of a triangle, with one orange in the gap between two oranges in the row below it, and one orange to start the arrangement. The number of rows in the triangle is double the number of layers in the pyramid. Find all possible values that the number of oranges can take.

Let the number of layers in the pyramid be n . The total number of oranges will be:

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Since the number of layers in the pyramid is n , the number of rows in the triangle will be double of that, which is $2n$.

And the total number of oranges must be:

$$1 + 2 + \dots + 2n = \frac{2n(2n+1)}{2}$$

But the total number of oranges in the pyramid is the same as the number of oranges in the triangle:

$$\frac{2n(2n+1)}{2} = \frac{n(n+1)(2n+1)}{6}$$

Multiply both sides by 6:

$$6n(2n+1) = n(n+1)(2n+1)$$

Divide both sides by $n(2n+1)$:

$$n+1 = 6 \Rightarrow n = 5$$

The total number of oranges can only be:

$$55$$

F. Cubes of natural numbers

1.49: Cube of the first n natural numbers

$$1^3 + 2^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Note the interesting property that the sum of the cubes of the first n natural numbers is the square of the sum of the first n natural numbers.

Example 1.50

Find the sum

- A. $1^3 + 2^3 + 3^3 + 4^3$
- B. $1^3 + 2^3 + \dots + 10^3$
- C. $5^3 + 6^3 + \dots + 10^3$

$$1^3 + 2^3 + 3^3 + 4^3 = \left[\frac{4(5)}{2} \right]^2 = 10^2 = 100$$

$$1^3 + 2^3 + \dots + 10^3 = \left[\frac{10(11)}{2} \right]^2 = 55^2 = 3025$$

$$5^3 + 6^3 + \dots + 10^3 = 3025 - 100 = 2925$$

Example 1.51

If $x, y \in \mathbb{N}$, then for what values of x and y does the inequality below hold true?

$$\left[\frac{x(x+1)}{2} \right]^2 < 1^3 + 2^3 + 3^3 + \dots + y^3$$

$$\left[\frac{x(x+1)}{2} \right]^2 < \left[\frac{y(y+1)}{2} \right]^2 \Rightarrow x < y$$

G. Removing Squares

Example 1.52

A set of tiles numbered 1 through 100 is modified repeatedly by the following operation: remove all tiles numbered with a perfect square, and renumber the remaining tiles consecutively starting with 1. How many times must the operation be performed to reduce the number of tiles in the set to one? (AMC 10A 2002/22)

Brute Force Method

Step I

We start with the tiles:

$$1, 2, \dots, 100$$

We remove the tiles

$$1^2, 2^2, \dots, 10^2 \Rightarrow 10 \text{ Tiles}$$

The number of tiles left is $100 - 10 = 90$. We renumber the remaining tiles, giving us:

$$1, 2, \dots, 90$$

Step II

We remove the tiles

$$1^2, 2^2, \dots, 9^2 \Rightarrow 9 \text{ Tiles}$$

The number of tiles left is $90 - 9 = 81$. We renumber the remaining tiles, giving us:

$$1, 2, \dots, 81$$

Simply calculate the number of steps required.

100	90	81	72	64	56	49	42	36	30	25	20	16	12	9	6	4	2	1
10	9	9	8	8	7	7	6	6	5	5	4	4	3	3	2	2	1	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	

Algebraic Method

We originally with a perfect square number of tiles. Let that perfect square be n^2 . After the first operation, we are left with

$$n^2 - n \text{ tiles}$$

$n^2 - n$ will have $n - 1$ perfect squares smaller than it, so subtract that:

$$n^2 - n - (n - 1) = n^2 - 2n + 1 = (n - 1)^2$$

Example 1.53: Continuation

Answer the question above if:

- there are 2500 tiles instead of 100?
- there are 2500 tiles instead of 100, and you stop when you reach 100 tiles instead of reaching 1 tile?
- if there are 104 tiles instead of 100

Part A

You want to go from

$$2500 = 50^2 \rightarrow 1^2 = 1 \Rightarrow 50 - 1 = 49$$

Number of steps needed

$$= 49 \times 2 = 98$$

Part B

You want to go from

$$2500 = 50^2 \rightarrow 100 = 10^2 \Rightarrow 50 - 10 = 40$$

Number of steps needed

$$= 40 \times 2 = 80$$

Part C

100	90	81	72	64	56	49	42	36	30	25	20	16	12	9	6	4	2	1
10	9	9	8	8	7	7	6	6	5	5	4	4	3	3	2	2	1	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	

104	94	85	76	68	60	53	46	40	34	29	24	20	16	12	9	6	4	2	1
10	9	9	8	8	7	7	6	6	5	5	4	4	4	3	3	2	2	1	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		

19 Steps

H. Number Theory

Example 1.54

In the sequence, 1, 22, 333,..., 10101010101010101010, 11111111111111111111, the sum of the digits in the 200th term is: (NMTC Primary/Screening 2004/03)

The pattern that we observe is:

1 repeats once

2 repeats twice

10 repeats ten times

11 repeats 11 times.

The n^{th} term is n repeated n times. Hence, the 200th term will consist of the number 200 repeated 200 times.
200200200...200

We need the sum of the digits of the above number.

The zeroes do not matter, because they will not increase the sum.

The 2 is repeated 200 times. Hence, the sum of digits is:

$$2 \times 200 = 400$$

Example 1.55

In the sequence of numbers 1, 2, 11, 22, 111, 222, ... the sum of digits in the 999th term is (NMTC Primary/Screening 2004/06)

The even numbered terms:

$$\begin{aligned} 2^{\text{nd}} \text{ Term: } & 1 \text{ Two} \\ 4^{\text{th}} \text{ Term: } & 2 \text{ Two's} \\ 6^{\text{th}} \text{ Term: } & 3 \text{ Two's} \\ n^{\text{th}} \text{ Term: } & \frac{n}{2} \text{ Two's} \end{aligned}$$

The odd numbered terms which are one less than the even numbered terms have the same number of digits:

$$\begin{aligned} 1000^{\text{th}} \text{ Term has } & \frac{1000}{2} = 500 \text{ Digits} \\ 999^{\text{th}} \text{ Term also has } & 500 \text{ Digits} \end{aligned}$$

$$\text{Sum of Digits} = \underbrace{1 + 1 + \dots + 1}_{500 \text{ Times}} = 1 \times 500 = 500$$

Example 1.56

A group of 12 pirates agree to divide a treasure chest of gold coins among themselves as follows. The k^{th} pirate to take a share takes $\frac{k}{12}$ of the coins that remain in the chest. The number of coins initially in the chest is the smallest number for which this arrangement will allow each pirate to receive a positive whole number of coins. How many coins does the 12th pirate receive? (AMC 10A 2013/21)

Let the chest initially have C coins. The first pirate takes $\frac{1}{12}$ of the coins, leaving

$$C \cdot \frac{11}{12} \text{ coins}$$

The second pirate takes $\frac{2}{12}$ of the remaining coins, leaving

$$C \cdot \frac{11}{12} \cdot \frac{10}{12} \text{ coins}$$

Continue this process, and the last pirate gets:

$$C \cdot \frac{11}{12} \cdot \frac{10}{12} \cdot \frac{9}{12} \cdot \frac{8}{12} \cdot \frac{7}{12} \cdot \frac{6}{12} \cdot \frac{5}{12} \cdot \frac{4}{12} \cdot \frac{3}{12} \cdot \frac{2}{12} \cdot \frac{1}{12}$$

Prime factor, write in exponent form and simplify:

$$C \cdot \frac{11 \times 7 \times 5^2 \times 3^4 \times 2^8}{12^{11}} = C \cdot \frac{11 \times 7 \times 5^2 \times 3^4 \times 2^8}{2^{22} \times 3^{11}} = C \cdot \frac{11 \times 7 \times 5^2}{2^{14} \times 3^7}$$

The smallest value that C can take is $2^{14} \times 3^7$.

The last pirate will get:

$$11 \times 7 \times 5^2 = 1925$$

I. Formula for Nth Term

Example 1.57

Consider the sequence

$$\frac{3}{5}, \frac{6}{7}, 1, 1, \frac{1}{11}$$

The 2016th term of the sequence is $\frac{p}{q}$ where p and q are integers having no common factors. The value of $q - p$ is: (NMTC Primary/Screening/2016/11)

Rewrite the sequence as:

$$\frac{3}{5}, \frac{6}{7}, \frac{9}{9}, \frac{12}{11}$$

Note that the numerators are consecutive multiples of 3, and denominators are odd numbers starting from 5. Hence, further rewrite the sequence as:

$$\frac{3 \cdot 1}{3 + 2 \cdot 1}, \frac{3 \cdot 2}{3 + 2 \cdot 2}, \frac{3 \cdot 3}{3 + 2 \cdot 3}, \frac{3 \cdot 4}{3 + 2 \cdot 4}$$

Note the bold violet number is also the term number. Hence, the general term is:

$$\frac{3n}{3 + 2n}$$

The 2016th term is then:

$$\frac{3 \cdot 2016}{3 + 2 \cdot 2016} = \frac{3 \cdot 2016}{4035} = \frac{2016}{1345} = \frac{p}{q}$$

Hence, the final answer is:

$$q - p = 1345 - 2016 = -671$$

Example 1.58: Exponents

- Find $S_{100} - S_{99}$ if $S_1 = 1, S_2 = 7, S_3 = 7^2, \dots$
- If $P_n = S_{n+1} - S_n$, then find a formula for P_n , and also find its first four terms.
- If $S_1 = 1, S_2 = x, S_3 = x^2, \dots$ then repeat parts A and B for the new sequence
- Write the sequence 5, 30, 180, 1080 in terms of the sequences given above.

Part A

The formula for the n^{th} term is 7^{n-1} :

$$S_{100} - S_{99} = 7^{99} - 7^{98} = 7^{98}(7 - 1) = 6 \times 7^{98}$$

Part B

$$P_n = S_{n+1} - S_n = 7^n - 7^{n-1} = 7^{n-1}(7 - 1) = 6 \times 7^{n-1}$$

$$P_1 = 7 - 1 = 6$$

$$P_2 = 49 - 7 = 42$$

$$P_3 = 343 - 49 = 294$$

$$P_4 = 2058$$

Part C

$$S_{100} - S_{99} = x^{99} - x^{98} = x^{98}(x - 1)$$

$$P_n = S_{n+1} - S_n = x^n - x^{n-1} = x^{n-1}(x - 1)$$

Part D

$$5 = 5 \times 1 = 6^1 - 6^0$$

$$30 = 5 \times 6 = 6^2 - 6^1$$

$$180 = 5 \times 6^2 = 6^3 - 6^2$$

$$1080 = 5 \times 6^3 = 6^4 - 6^3$$

1.4 Pairing and Factoring

A. Pairing

Sometimes terms do not cancel. Rather, combining terms makes the calculations simple. Being able to see which terms to combine requires practice.

Example 1.59

- A. Find the value of: $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8$
- B. Find the value of: $1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9$
- C. Find the value of: $1 - 2 + 3 - 4 + \dots + (n - 1) - n$, n is even
- D. Find the value of: $1 - 2 + 3 - 4 + \dots - (n - 1) + n$, n is odd

Part A

Make a pair of two consecutive numbers:

$$\underbrace{1-2}_{1-2=-1} + \underbrace{3-4}_{3-4=-1} + \underbrace{5-6}_{5-6=-1} + \underbrace{7-8}_{7-8=-1} = -1 - 1 - 1 - 1 = -4$$

Part B

$$\underbrace{1-2}_{1-2=-1} + \underbrace{3-4}_{3-4=-1} + \underbrace{5-6}_{5-6=-1} + \underbrace{7-8}_{7-8=-1} + 9 = -1 - 1 - 1 - 1 + 9 = 5$$

You can also do this by pairing the first term with the last term, and so on, leaving the middle term

$$1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + 9 = 5$$

Part C

The answer to this question is the same as the answer to Part A, except that there we had four pairs, and hence we multiplied -1 by 4.

Here, we have n numbers, and hence we will have $\frac{n}{2}$ pairs, giving us:

$$\underbrace{1-2}_{1-2=-1} + \underbrace{3-4}_{3-4=-1} + \dots + \underbrace{(n-1)-n}_{(n-1)-1=-1} = \underbrace{-1 - 1 - \dots - 1}_{\frac{n}{2} \text{ Times } -1} = -\frac{n}{2}$$

Part D

Method I: Make pairs of all except the last number

$$\underbrace{1-2}_{1-2=-1} + \underbrace{3-4}_{3-4=-1} + \dots + \underbrace{(n-2)-(n-1)}_{(n-2)-(n-1)=-1} + n$$

Each pair adds up -1 giving us:

$$\underbrace{-1 - 1 - \dots - 1}_{\frac{n-1}{2} \text{ Times } -1} + n = -\frac{(n-1)}{2} + n = \frac{n+1}{2}$$

Method II: Make pairs of all the numbers

$$1 - 2 + 3 - 4 + \dots - (n-1) + n$$

Add the pairs:

$$\begin{aligned} 1 + n &= n + 1 \\ -2 - (n-1) &= -2 - n = -(n+1) \end{aligned}$$

All the pairs will cancel, except for the middle number, which will be

$$\frac{n+1}{2}$$

Example 1.60

Saket added up all the even numbers from 1 to 101. Then, from the total he obtained, he subtracted all odd numbers between 0 and 100. The answer he would have obtained is: (NMTCT Primary-III/Screening/49)

$$\left(\underbrace{2 + 4 + \dots + 100}_{\text{Even Numbers}} \right) - \left(\underbrace{1 + 3 + \dots + 99}_{\text{Odd Numbers}} \right)$$

Rearrange:

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$$\underbrace{(2 - 1) + (4 - 3) + \cdots + (100 - 99)}_{50 \text{ Pairs}}$$

Note that every pair simplifies to 1:

$$\underbrace{1 + 1 + \cdots + 1}_{50 \text{ Times}} = 50$$

Example 1.61

Asha adds all the odd integers from 1 through 101, inclusive, and then subtracts all the even integers in that same range from her sum. What result does she obtain? (**MathCounts 2007 State Countdown**)

$$\begin{aligned} & (1+3+\dots+101)-(2+4+\dots+100) \\ & = (2-1)+(4-3)+\dots+(100-99)+101 \\ & = -50+101 \\ & = 51 \end{aligned}$$

Example 1.62

What is the positive difference between the sum of the first 20 positive even integers and the sum of the first 15 positive odd integers? (**MathCounts 2001 National Sprint**)

$$\underbrace{(2 + 4 + \cdots + 40)}_{\text{Even Numbers}} - \underbrace{(1 + 3 + \cdots + 29)}_{\text{Odd Numbers}}$$

Rearrange:

$$\underbrace{(2 - 1) + (4 - 3) + \cdots + (30 - 29)}_{15 \text{ Pairs}} + 32 + 34 + 36 + 38 + 40$$

Simplify:

$$\underbrace{1 + 1 + \cdots + 1}_{15 \text{ Times}} + 36 + 36 + 36 + 36 + 36 = 15 + 36 \times 5 = 195$$

Example 1.63

Find the sum of the largest prime factor and the smallest prime factor of $(-498) + (-497) + \cdots + 500$.

$$(-\mathbf{498}) + (-\mathbf{497}) + \cdots + \mathbf{497} + \mathbf{498} + 499 + 500$$

All the pairs will cancel out:

$$499 + 500 = 999$$

Prime Factorize:

$$999 = 3^3 \times 37 \Rightarrow 3 + 37 = 40$$

Example 1.64

What is the difference between the sum of the first 2003 even counting numbers and the sum of the first 2003 odd counting numbers? (**AMC 10A 2003/1, AMC 12A 2003/1**)

$$\underbrace{2 + 4 + \cdots}_{2003 \text{ Numbers}} - \underbrace{(1 + 3 + \cdots)}_{2003 \text{ Numbers}}$$

Rearrange:

$$= \underbrace{(2 - 1) + (4 - 3) + \cdots}_{2003 \text{ Pairs}}$$

Each pair simplifies to 1. Since there are 2003 such pairs, the total will be $1 \times 2003 = 2003$

Example 1.65

(MathCounts Chapter Sprint 2022/22)

Alva adds up all the positive integers less than 2023 and Beth adds up all the even positive integers less than 2023. What is the remainder when the absolute difference between Alva's and Beth's sums is divided by 1000?

$$(1 + 2 + 3 + \cdots + 2022) - (2 + 4 + \cdots + 2022)$$

We end up subtracting all the even numbers, leaving us with all the odd numbers upto 2021:
 $= 1 + 3 + 5 + \cdots + 2021$

$$\begin{aligned} 1 &= 2(1) - 1 \\ 3 &= 2(2) - 1 \\ 5 &= 2(3) - 1 \\ &\vdots \\ &\vdots \\ &\vdots \\ 2021 &= 2(1011) - 1 \end{aligned}$$

Hence, we want the sum of the first 1011 odd numbers, which using the property

$$1 + 3 + 5 + \cdots + n^{\text{th}} \text{ odd number} = n^2$$

Becomes

$$= 1011^2$$

The remainder on dividing by 1000 gives us the last three digits. Hence, we can drop the thousand's digit from the calculations:

$$= 11^2 = 121$$

Example 1.66

The terms $x, x + 2, x + 4, \dots, x + 2n$ form an arithmetic sequence, with x an integer. If each term of the sequence is cubed, the sum of the cubes is -1197 . What is the value of n if $n > 3$? (MathCounts 2004 State Sprint)

x is odd since the final answer is odd.

$$\begin{aligned} (-1)^3 + (-3)^3 + \cdots + (-9)^3 &= -1225 \\ -1125 - (-1197) &= -28 = (-1)^3 + (-3)^3 \\ -9, -7, -5 &\Rightarrow n = 2 \\ -9, -7, -5, -3, -1, 1, 3 &\Rightarrow n = 6 \end{aligned}$$

Example 1.67

What is the greatest number of consecutive integers whose sum is 45? (AMC 10A 2019/5, AMC 12A 2019/4)

$$-44, -43, \dots, 43, 44, 45 \Rightarrow 90 \text{ Numbers}$$

Example 1.68

If $S_n = \underbrace{1 - 2 + 3 - 4 + \cdots}_{n \text{ terms}}$, then find:

- A. $S_{2021} + S_{2022}$
- B. $S_n + S_{n+1}$

Part A

$$S_{2022} = -\frac{2022}{2} = -1011$$

$$S_{2021} = \frac{2021+1}{2} = \frac{2022}{2} = 1011$$

$$S_{2021} + S_{2022} = 0$$

Part B

$$n \text{ is even: } S_n + S_{n+1} = -\frac{n}{2} + \frac{(n+1)+1}{2} = \frac{-n+n+2}{2} = \frac{2}{2} = 1$$

$$n \text{ is odd: } S_n + S_{n+1} = \frac{n+1}{2} + \frac{-(n+1)}{2} = \frac{0}{2} = 0$$

Example 1.69

Consider the sequence

$$1, -2, 3, -4, 5, -6, \dots$$

whose n^{th} term is $(-1)^{n+1} \cdot n$. What is the average of the first 200 terms of the sequence? (AHSME 1997/6)

Example 1.70

Daryl first writes the perfect squares as a sequence

$$1, 4, 9, 16, 25, 36, 49, 64, 81, 100, \dots$$

After the number 1, he then alternates by making two terms negative followed by leaving two terms positive. Daryl's new sequence is

$$1, -4, -9, 16, 25, -36, -49, 64, 81, -100, \dots$$

What is the sum of the first 2011 terms in this new sequence? (CEMC Grade 8 2011/25)

Consider groups of four consecutive terms in the sequence, starting from the first. They can be written:

$$n^2 - (n+1)^2 - (n+2)^2 + (n+3)^2$$

Re-arrange:

$$(n+3)^2 + n^2 - (n+1)^2 - (n+2)^2$$

Expand and simplify:

$$\begin{aligned} &= n^2 + 6n + 9 + n^2 - [n^2 + 2n + 1 + n^2 + 4n + 4] \\ &= 2n^2 + 6n + 9 - [2n^2 + 6n + 5] \\ &= 4 \end{aligned}$$

Hence, every set of 4 terms adds up to 4.

We have 2011 terms. We use complementary counting and find the total for 2012 terms, and then adjust it for the missing term.

$$= 2012 \text{ terms} = 503 \text{ groups} \Rightarrow \text{Total} = 503 \times 4 = 2012$$

But, the 2012^{th} term is missing, and that is:

$$2012^2 = 4,048,144$$

The final answer is:

$$2012 - 4048144 = 4046132$$

B. Factoring

Example 1.71

You are given a sequence of 58 terms; each term has the form $P + n$ where P stands for the product $2 \times 3 \times 5 \times \dots \times 61$ of all prime numbers less than or equal to 61, and n takes, successively, the values 2, 3, 4, ..., 59. Let N be the number of primes appearing in this sequence. Then N is: (AHSME 1960/)

For each value of the sequence, we can take some prime p common:

$$P + n = p \left(\frac{P}{n} + 1 \right)$$

Hence, the number of primes is zero.

1.5 Telescoping

A. Products

1.72: Telescope

We can sometimes write out a series in such a way that adjacent terms start cancelling. Telescoping products use cancellation in the numerator and the denominator.

Example 1.73: Products

- A. $\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5}$
 B. $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6} \times \frac{6}{7} \times \frac{7}{8} \times \frac{8}{9} \times \frac{9}{10}$
 C. $\frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{99}{100}$
 D. $\frac{a}{a+1} \times \frac{a+1}{a+2} \times \dots \times \frac{a+n-1}{a+n}$
 E. If $\frac{4}{5} \times \frac{5}{6} \times \dots \times \frac{a}{a+1} = \frac{1}{n}$, $n \in \mathbb{N}$, then find the possible values of a .

Part A

Note that the denominator and the numerator of successive terms cancels, leaving us with only the numerator of the first term, and the denominator of the last term:

$$\frac{2}{\cancel{3}} \times \frac{\cancel{3}}{\cancel{4}} \times \frac{\cancel{4}}{5} = \frac{2}{1} \times \frac{1}{1} \times \frac{1}{5} = \frac{2}{5}$$

Part B

The idea here is the same as in Part A. We are again left with the numerator of the first term, and the denominator of the last term:

$$\frac{2}{\cancel{3}} \times \frac{\cancel{3}}{\cancel{4}} \times \dots \times \frac{\cancel{99}}{100} = \frac{2}{100} = \frac{1}{50}$$

Part C

$$\frac{a}{\cancel{a+1}} \times \frac{\cancel{a+1}}{\cancel{a+2}} \times \dots \times \frac{\cancel{a+n-1}}{a+n} = \frac{a}{a+n}$$

Part D

$$\frac{4}{5} \times \frac{5}{6} \times \dots \times \frac{a}{a+1} = \frac{4}{a+1} = \frac{1}{n}$$

$$4n = a + 1 \Rightarrow a = 4n - 1, n \geq 2$$

For example:

$$n = 2 \Rightarrow a = 4(2) - 1 = 8 - 1 = 7 \Rightarrow a + 1 = 8 \Rightarrow \frac{4}{8} = \frac{1}{2}$$

However, $n = 1$ is not possible since

$$a = 4(1) - 1 = 4 - 1 = 3 \Rightarrow \text{This is less}$$

Example 1.74

(NMTC Primary/2018/17)

17. The value of $\left(1 + \frac{1}{9}\right)\left(1 + \frac{1}{8}\right)\left(1 + \frac{1}{7}\right)\left(1 + \frac{1}{6}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{2}\right)$ is _____

$$\begin{aligned} & \left(1 + \frac{1}{9}\right)\left(1 + \frac{1}{8}\right)\left(1 + \frac{1}{7}\right)\left(1 + \frac{1}{6}\right)\left(1 + \frac{1}{5}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{2}\right) \\ &= \left(\frac{10}{9}\right)\left(\frac{9}{8}\right)\left(\frac{8}{7}\right)\left(\frac{7}{6}\right)\left(\frac{6}{5}\right)\left(\frac{5}{4}\right)\left(\frac{4}{3}\right)\left(\frac{3}{2}\right) \\ &= 5 \end{aligned}$$

Example 1.75: Setting up a Telescope

- A. $\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)$ (CEMC Fermat 2012/4)
- B. Evaluate $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{6}\right)\left(1 - \frac{1}{7}\right)$
- C. The product of the 9 factors $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{10}\right) =$ (AMC 8 1985/9)
- D. Evaluate $\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{6}\right) \cdots \left(1 - \frac{1}{23}\right)\left(1 - \frac{1}{24}\right)$
- E. Evaluate $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{n-1}\right)\left(1 - \frac{1}{n}\right)$
- F. If $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{n}\right) = \frac{1}{32}$, then $n = ?$
- G. $\sqrt{\frac{3}{1} \cdot \frac{5}{3} \cdot \frac{7}{5} \cdots \frac{2n+1}{2n-1}} = 9$ (CEMC Fermat 2002/21)

$$\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right) = \left(\frac{4}{3}\right)\left(\frac{5}{4}\right) = \frac{5}{3}$$

Part A

$$1 - \frac{1}{2} = \frac{2}{2} - \frac{1}{2} = \frac{1}{2}$$

Evaluate each term. Then, do not multiply! See that the terms cancel.

$$\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right)\left(\frac{4}{5}\right)\left(\frac{5}{6}\right)\left(\frac{6}{7}\right) = \frac{1}{7}$$

Part B

$$\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) \cdots \left(\frac{9}{10}\right) = \frac{1}{10}$$

Part C

Use three dots to indicate that the pattern continues (rather than writing out the whole sum). All terms except the first numerator and the last denominator will cancel.

$$\left(\frac{3}{4}\right)\left(\frac{4}{5}\right)\left(\frac{5}{6}\right) \dots \left(\frac{23}{24}\right) = \frac{3}{24} = \frac{1}{8}$$

Part D

This question has n instead of the last denominator, but the same pattern:

$$\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{3}{4}\right) \dots \left(\frac{n-1}{n}\right) = \frac{1}{n}$$

Part E

From the previous example, we see that the denominator of the last fraction is the denominator of the answer.
 Hence, $n = 32$

Part G

Square both sides:

$$\frac{3}{1} \cdot \frac{5}{3} \cdot \frac{7}{5} \dots \frac{2n+1}{2n-1} = 81$$

Telescope:

$$2n + 1 = 81 \Rightarrow 2n = 80 \Rightarrow n = 40$$

Example 1.76

One half of the water is poured out of a full container. Then one third of the remainder is poured out. Continue the process: one fourth of the remainder for the third pouring, one fifth of the remainder for the fourth pouring, etc. After how many pourings does exactly one tenth of the original water remain? (AMC 8 1992/25)

$$\left(\frac{1}{2}\right)\left(\frac{2}{3}\right) \dots \left(\frac{9}{10}\right) = \frac{1}{10} \Rightarrow \text{No. of Terms} = 9 \Rightarrow 9 \text{ Pourings}$$

Example 1.77

Let O_n be the sum of the first n positive odd integers, and let E_n be the sum of the first n positive even integers. For example, $E_1 = 2$, and $O_2 = 1 + 3 = 4$. What is the value of the product (BmMT 2024/Individual/9)

$$\left(\frac{O_1}{E_1}\right)\left(\frac{O_2}{E_2}\right) \dots \left(\frac{O_{2024}}{E_{2024}}\right)$$

The general term is:

$$= \frac{O_n}{E_n} = \frac{n^2}{n(n+1)} = \frac{n}{n+1}$$

Hence, the product becomes:

$$\left(\frac{1}{2}\right)\left(\frac{2}{3}\right) \dots \left(\frac{2024}{2025}\right) = \frac{1}{2025}$$

B. Sums

1.78: Breaking $\frac{1}{n(n+1)}$

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1}{n(n+1)} - \frac{n}{n(n+1)} = \frac{1}{n(n+1)}$$

Take the LCM

$$\frac{1}{2} - \frac{1}{3} = \frac{3}{6} - \frac{2}{6} = \frac{1}{6} = \frac{1}{2 \times 3}$$

$$\frac{1}{4} - \frac{1}{5} = \frac{5}{20} - \frac{4}{20} = \frac{1}{20} = \frac{1}{4 \times 5}$$

Example 1.79

- A. The value of $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \frac{1}{7 \times 8}$ (NMTC Primary/Screening 2007/13)
- B. Evaluate $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30}$
- C. Evaluate $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \dots + \frac{1}{n(n-1)}$

Part A

Apply the formula to break each term into two:

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{7} - \frac{1}{8}\right) = 1 - \frac{1}{8} = \frac{7}{8}$$

Part B

Rewrite the denominators:

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6}$$

Apply the formula to break each term into two:

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right)$$

Everything cancels, except the first and the last term:

$$1 - \frac{1}{6} = \frac{5}{6}$$

Part C

Rewrite the denominators:

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{(n-1)(n-2)} + \frac{1}{n(n-1)}$$

Apply the formula to break each term into two:

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n-2} - \frac{1}{n-1}\right) + \left(\frac{1}{n-1} - \frac{1}{n}\right)$$

Everything cancels, except the first and the last term:

$$= 1 - \frac{1}{n} = \frac{n-1}{n}$$

Example 1.80

NMTC Primary/2015/13

13. The value of $\frac{50}{72} + \frac{50}{90} + \frac{50}{110} + \dots + \frac{50}{9900}$ is

a) $\frac{23}{4}$

b) $\frac{32}{7}$

c) $\frac{1}{2015}$

d) $\frac{55}{27}$

$$\frac{50}{72} + \frac{50}{90} + \frac{50}{110} + \dots + \frac{50}{9900}$$

$$= 50 \left(\frac{1}{8 \times 9} + \frac{1}{9 \times 10} + \frac{1}{10 \times 11} + \dots + \frac{1}{99 \times 100} \right)$$

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Aziz Manva (azizmanva@gmail.com)

$$\begin{aligned} &= 50 \left[\left(\frac{1}{8} - \frac{1}{9} \right) + \left(\frac{1}{9} - \frac{1}{10} \right) + \left(\frac{1}{10} - \frac{1}{11} \right) + \cdots + \left(\frac{1}{99} - \frac{1}{100} \right) \right] \\ &= 50 \left[\frac{1}{8} - \frac{1}{100} \right] = \frac{50}{8} - \frac{50}{100} = \frac{25}{4} - \frac{1}{2} = \frac{25}{4} - \frac{2}{4} = \frac{23}{4} \end{aligned}$$

1.6 Further Topics

81 Examples